

OULU VIHREASAARI, Finland

WMO: 028760

Lat: 65.006N

Lon: 25.393E

Elev: 4

StdP: 101.28

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.5	-22.3	-28.3	0.3	-25.5	-24.7	0.4	-22.2	15.6	1.2	13.3	-1.3	2.2	140	0.759

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.4	24.5	18.8	22.6	18.1	21.0	17.1	20.1	23.4	18.8	21.7	17.7	20.5	4.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.6	22.0	17.5	12.5	20.7	16.4	11.6	19.5	57.6	23.5	53.3	21.6	49.8	20.5	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
13.2	11.3	10.0	DB	-26.3	26.7	5.0	3.0	-29.9	28.9	-32.9	30.6	-35.7	32.3	-39.4	34.4	(4)
			WB	-26.4	20.9	4.7	2.1	-29.8	22.5	-32.6	23.7	-35.2	24.9	-38.6	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-9.2	-7.7	-4.7	0.9	8.3	13.2	17.1	15.1	10.7	4.1	-0.4	-4.5
	DBStd	9.96	6.80	7.22	5.44	3.35	4.52	3.53	3.25	2.84	2.98	3.31	4.14	5.97
	HDD10.0	2887	594	496	456	273	87	9	0	1	27	184	311	449
	HDD18.3	5396	852	729	714	523	311	158	63	106	228	442	561	707
	CDD10.0	569	0	0	0	0	35	106	219	160	49	1	0	0
	CDD18.3	38	0	0	0	0	2	5	24	7	0	0	0	0
	CDH23.3	126	0	0	0	0	9	15	84	18	0	0	0	0
	CDH26.7	22	0	0	0	0	0	3	18	1	0	0	0	0
Wind	WSAvg	5.0	4.7	5.0	5.0	4.4	5.0	5.3	4.4	4.7	5.1	5.5	5.4	5.3
Precipitation	PrecAvg	596	49	40	35	32	42	53	73	61	54	55	57	52
	PrecMax	802	82	82	56	75	81	102	133	142	96	92	96	94
	PrecMin	451	17	6	12	10	9	7	11	8	9	16	13	14
	PrecStd	111	18	20	13	17	19	24	32	34	26	22	24	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	4.5	5.7	11.9	24.8	25.4	29.0	25.9	19.4	11.9	8.3	3.7
		MCWB	1.7	2.8	3.0	8.0	14.9	18.5	19.6	20.1	15.4	10.0	6.9	2.7
	2%	DB	1.4	2.9	3.7	8.2	20.0	22.1	25.3	22.6	17.1	10.5	6.8	3.0
		MCWB	0.7	1.7	1.8	5.1	13.5	16.5	19.5	18.6	14.1	9.2	5.9	2.0
	5%	DB	0.6	1.7	2.7	6.1	17.0	20.0	23.4	20.7	15.7	9.5	5.5	2.3
		MCWB	0.2	0.8	1.2	3.7	12.3	15.8	18.9	17.2	13.5	8.2	4.7	1.5
	10%	DB	-0.3	0.7	1.8	4.9	14.3	18.1	21.9	19.3	14.8	8.5	4.2	1.6
		MCWB	-0.7	0.1	0.7	3.1	10.6	14.6	18.0	16.3	12.8	7.3	3.6	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	2.9	3.3	9.1	15.7	19.1	22.0	21.0	16.1	11.0	7.1	3.1
		MCDB	2.7	4.0	5.1	11.4	21.7	22.5	24.9	24.1	18.7	11.4	7.9	3.5
	2%	WB	0.9	1.9	2.2	5.5	14.0	17.8	20.8	19.3	14.7	9.3	6.0	2.2
		MCDB	1.2	2.9	3.3	8.0	18.7	21.1	24.0	22.3	16.5	10.2	6.8	2.8
	5%	WB	0.3	1.0	1.5	4.2	12.5	16.5	19.6	17.8	13.9	8.4	4.9	1.6
		MCDB	0.7	1.6	2.5	5.6	16.7	19.4	22.7	20.1	15.4	9.4	5.4	2.1
	10%	WB	-0.8	0.3	0.8	3.3	11.0	15.0	18.5	16.7	13.1	7.5	3.7	1.0
		MCDB	-0.3	0.9	1.7	4.6	13.7	17.8	21.4	18.7	14.6	8.4	4.1	1.5
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.0	6.7	5.4	5.8	5.1	5.4	5.2	4.6	3.7	3.3	4.3
		MCDBR	4.9	4.3	5.1	6.8	10.0	8.1	7.9	7.1	6.0	4.0	3.8	3.5
		MCWBR	4.8	3.9	4.1	4.4	5.2	4.5	4.7	4.3	3.8	3.3	3.4	3.1
	5% WB	MCDBR	4.9	3.7	4.6	6.1	9.0	7.1	7.0	5.9	5.2	4.0	3.5	3.3
		MCWBR	4.8	3.5	3.8	4.4	5.3	4.4	4.6	4.1	3.6	3.5	3.3	3.0
Clear-Sky Solar Irradiance	taub	0.209	0.278	0.309	0.346	0.348	0.351	0.370	0.354	0.331	0.307	0.344	0.355	
	taud	1.952	2.108	2.161	2.241	2.425	2.468	2.436	2.473	2.519	2.356	2.551	2.606	
	Ebn at Noon	294	596	756	818	860	865	832	813	755	579	271	115	
	Edh at Noon	23	60	89	106	97	96	96	84	65	48	21	12	
All-Sky Solar Radiation	RadAvg	0.11	0.62	2.09	3.79	4.96	5.56	5.13	3.80	2.08	0.76	0.16	0.03	
	RadStd	0.01	0.12	0.21	0.30	0.47	0.44	0.46	0.36	0.21	0.12	0.03	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page